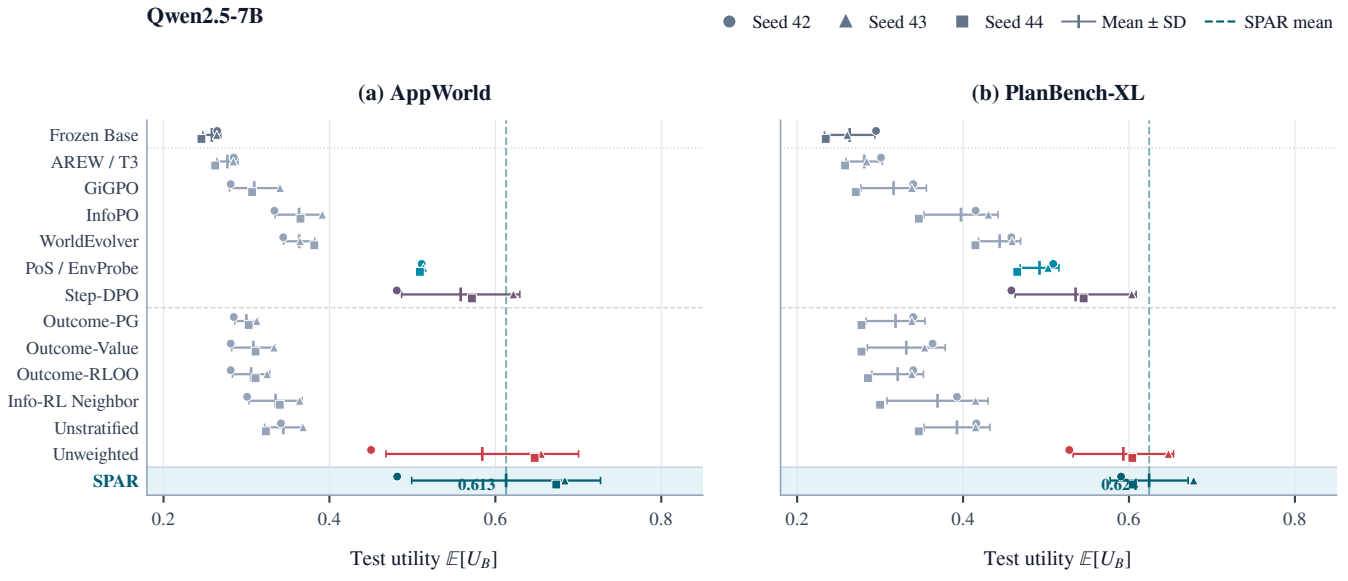
Qwen2.5-7B
Seed 42
Seed 43
Seed 44
Mean ± SD
SPAR mean
(a) AppWorld
(b) PlanBench-XL
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.613
0.624
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$
Test utility $\mathbb{E}[U_B]$